A Statistical Analysis of Cardiovascular Disease Risk Factors Using Minitab: A Gender-Based Comparative Approach

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A Comprehensive Project Report

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Abstract

Cardiovascular disease (CVD) remains a major cause of morbidity and mortality and is strongly associated with modifiable metabolic and behavioral factors. This project examined cardiovascular risk patterns in a large public dataset of 68,672 adults and evaluated whether the strength of those relationships differed between men and women. Analyses were conducted in Minitab 21 using Spearman rank correlation, Welch's two-sample t test, Kruskal–Wallis tests, a chi-square test of independence, binary logistic regression, and receiver operating characteristic (ROC) analysis.

Individuals with CVD had substantially higher mean body mass index (BMI) than those without CVD (28.45 vs. 26.47; $p < 0.001$). Systolic blood pressure differed significantly across glucose categories, and BMI increased across cholesterol categories. Cholesterol was strongly associated with CVD status ($\chi^2(2) = 3368.72$, $p < 0.001$). In the multivariable logistic model, age, BMI, systolic and diastolic blood pressure, and elevated cholesterol were important predictors of CVD, while physical activity was associated with lower adjusted odds. The combined model had an AUC of 0.791. Gender-specific models showed broadly similar risk structures, with somewhat stronger age effects among women and stronger BMI and physical-activity effects among men. Overall, the results support the use of multivariable statistical modeling for cardiovascular risk assessment while showing that the magnitude of selected associations can vary by sex.

1 Introduction

Cardiovascular diseases are a leading cause of death worldwide and remain a major public-health challenge in the United States (WHO, 2025; Roth et al., 2023; CDC, 2024). Much of the burden is linked to established and potentially modifiable risk factors, including hypertension, excess body weight, abnormal cholesterol, smoking, poor diet, diabetes, and physical inactivity. Because many of these factors occur together, multivariable statistical methods are useful for distinguishing their independent associations with disease.

Risk profiles may also differ between men and women. Prior research has reported sex-related differences in the timing of CVD onset and in the distribution of hypertension, obesity, smoking, and lipid abnormalities (Li et al., 2022). A practical statistical question is therefore not only which variables are associated with CVD, but whether their relative importance changes when models are estimated separately by gender.

This study uses a large public cardiovascular-health dataset to demonstrate an applied statistical workflow in Minitab. The emphasis is on interpretable methods that move from descriptive and group-comparison analyses to multivariable prediction and model-performance assessment.

1.1 Research Questions

The analysis addresses four questions:

1. How are age, BMI, blood pressure, and weight related to one another?
2. Do cardiovascular risk indicators differ between people with and without CVD or across glucose and cholesterol groups?
3. Which variables remain associated with CVD after simultaneous adjustment in a logistic regression model?
4. Do the estimated relationships differ meaningfully between women and men, and how well do the models discriminate CVD status?

2 Data and Methods

2.1 Data Source and Analytic Sample

The analysis used the publicly available Kaggle Cardiovascular Disease dataset, containing anonymized demographic, clinical, and lifestyle information for adults approximately 30–70 years of age (Kaggle, 2023). After cleaning and removal of implausible values, the analytic sample contained 68,672 observations. All statistical analyses were performed in Minitab 21.

Age was converted from days to years. BMI was computed as

$$\text{BMI} = \frac{\text{weight (kg)}}{[\text{height (m)}]^2}. \tag{1}$$

The source analysis retained observations with systolic blood pressure between 80 and 180 mmHg, diastolic blood pressure between 40 and 120 mmHg, and BMI between 10 and 50.

Table 1: Variables used in the analysis

Variable	Type	Interpretation
CVD status	Binary	1 = cardiovascular disease present; 0 = absent
Age	Continuous	Age in years
Gender	Categorical	Female or male
BMI	Continuous	Body mass index derived from height and weight
Systolic BP	Continuous	Systolic blood pressure (mmHg)
Diastolic BP	Continuous	Diastolic blood pressure (mmHg)
Cholesterol	Ordinal	Normal, above normal, or high
Glucose	Ordinal	Normal, above normal, or high
Smoking	Binary	Current smoking indicator
Alcohol	Binary	Alcohol-use indicator
Physical activity	Binary	Physical-activity indicator

2.2 Statistical Analysis

Because several continuous variables were non-normal, Spearman rank correlation was used to summarize monotonic relationships. Welch's two-sample t test compared mean BMI between CVD and non-CVD groups; the very large sample size makes the comparison of means robust to moderate departures from normality. Kruskal–Wallis tests compared systolic blood pressure across glucose categories and BMI across cholesterol categories. A chi-square test examined the association between CVD status and cholesterol category (Conover, 2011).

Binary logistic regression modeled

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k, \tag{2}$$

where $p = P(\text{CVD} = 1)$. Odds ratios were calculated as $\exp(\beta)$. A combined model was estimated first, followed by separate models for women and men. Predictive discrimination was summarized using ROC curves and the area under the curve (AUC).

3 Results

3.1 Correlation Structure

The strongest Spearman association was between BMI and weight ($r_s \approx 0.85$), which is expected because weight is part of the BMI calculation. Systolic and diastolic blood pressure were also strongly related ($r_s \approx 0.70$). Age had weaker positive associations with systolic blood pressure ($r_s \approx 0.21$), diastolic blood pressure ($r_s \approx 0.16$), and BMI ($r_s \approx 0.09$). These results suggest that most predictors provide distinct information, although BMI and weight are sufficiently redundant that only one should enter the multivariable model.

Table 2: Spearman correlation matrix among continuous variables

	Age	BMI	Systolic BP	Diastolic BP	Weight
Age	1.00	0.09	0.21	0.16	0.05
BMI	0.09	1.00	0.26	0.21	0.85
Systolic BP	0.21	0.26	1.00	0.70	0.27
Diastolic BP	0.16	0.21	0.70	1.00	0.24
Weight	0.05	0.85	0.27	0.24	1.00

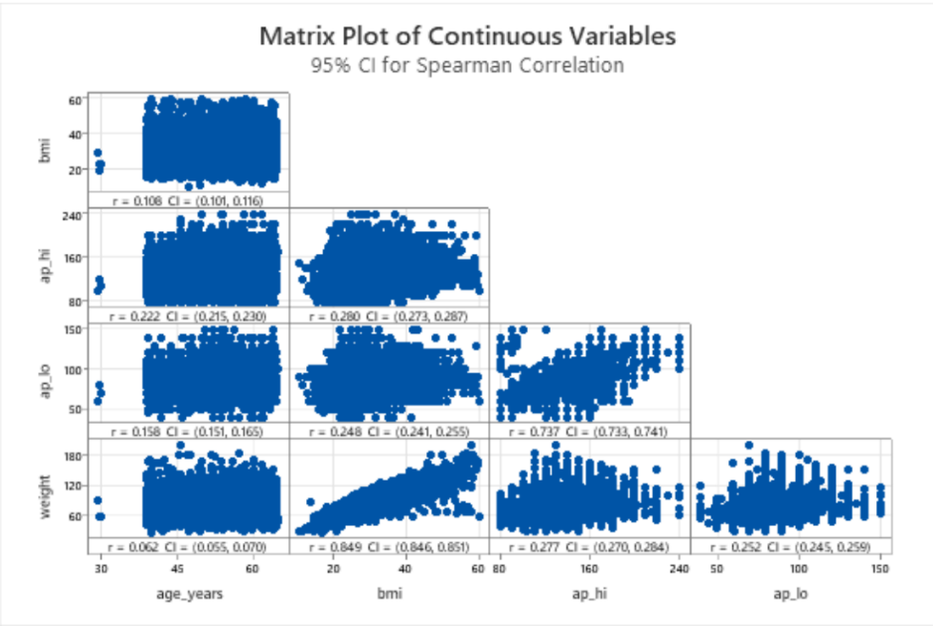


Figure 4.1: Spearman Correlation Matrix Plot

Interpretation of Spearman Correlation Results

Figure 1: Minitab matrix plot of the main continuous variables with Spearman correlation summaries.

3.2 BMI and CVD Status

Mean BMI was 26.47 (SD 4.77) among participants without CVD and 28.45 (SD 5.44) among those with CVD. Welch's t test estimated the mean difference (no CVD minus CVD) as -1.99 BMI units, with a 95% confidence interval from -2.07 to -1.91 ($t = -50.89$, $p < 0.001$). The result provides strong evidence that people with CVD had higher average BMI in this sample.

Table 3: BMI by cardiovascular disease status

CVD status	n	Mean	SD	SE
No CVD	34,685	26.47	4.77	0.026
CVD	33,987	28.45	5.44	0.030

3.3 Nonparametric Group Comparisons

Systolic blood pressure differed across glucose categories. Median systolic pressure was 120 mmHg in the normal-glucose group and 130 mmHg in both elevated-glucose groups. The tie-adjusted Kruskal–Wallis statistic was $H = 777.13$ ($p < 0.001$).

BMI also differed strongly across cholesterol categories. Median BMI increased from 25.87 in the normal-cholesterol group to 27.68 in the above-normal group and 28.65 in the high-cholesterol group. The tie-adjusted Kruskal–Wallis statistic was $H = 2086.67$ ($p < 0.001$). These results show that less favorable metabolic categories were accompanied by higher blood pressure or BMI distributions.

Table 4: Selected Kruskal–Wallis comparisons

Outcome and grouping variable	Group medians	H	p
Systolic BP by glucose category	120, 130, 130	777.13	< 0.001
BMI by cholesterol category	25.87, 27.68, 28.65	2086.67	< 0.001

3.4 CVD and Cholesterol

Cholesterol category was strongly associated with CVD status, $\chi^2(2) = 3368.72$, $p < 0.001$. Participants without CVD were observed more often than expected in the normal-cholesterol category, whereas participants with CVD were observed more often than expected in the above-normal and high categories. The standardized residual for high cholesterol was 33.80 among participants with CVD and -33.46 among those without CVD, indicating that this category contributed substantially to the overall association.

Table 5: Observed counts for CVD status and cholesterol category

	Normal	Above normal	High	Total
No CVD	29,062	3,755	1,868	34,685
CVD	22,432	5,546	6,009	33,987
Total	51,494	9,301	7,877	68,672

3.5 Multivariable Logistic Regression

The combined logistic model was statistically significant overall (Wald $\chi^2(12) = 12262.04$, $p < 0.001$). Age, BMI, systolic and diastolic blood pressure, and elevated cholesterol had positive adjusted coefficients. The largest categorical effect was high cholesterol: compared with normal cholesterol, the adjusted odds ratio was 2.98 (95% CI 2.78–3.19). A one-year increase in age was associated with approximately 5.2% higher adjusted odds of CVD, while a one-unit increase in BMI was associated with approximately 2.8% higher odds.

Physical activity was associated with lower adjusted odds (OR 0.79, 95% CI 0.76–0.83). Gender was not statistically significant in the combined model ($p = 0.108$). The source model also produced inverse adjusted coefficients for smoking, alcohol use, and the highest glucose category. Because these directions are counterintuitive and the project is observational, they should be

interpreted cautiously as conditional associations that may reflect coding, selection, treatment, or residual confounding rather than protective causal effects.

Table 6: Selected coefficients from the combined logistic regression model

Predictor	Coefficient	OR	95% CI for OR	p
Age (years)	0.05076	1.052	(1.049, 1.055)	< 0.001
BMI	0.02787	1.028	(1.025, 1.032)	< 0.001
Systolic BP	0.05369	1.055	(1.053, 1.057)	< 0.001
Diastolic BP	0.01593	1.016	(1.013, 1.019)	< 0.001
Cholesterol: above normal	0.3788	1.461	(1.384, 1.541)	< 0.001
Cholesterol: high	1.0907	2.976	(2.775, 3.192)	< 0.001
Physical activity	-0.2304	0.794	(0.761, 0.829)	< 0.001
Gender: male	0.0315	1.032	(0.993, 1.073)	0.108

The AUC for the combined model was 0.791, indicating useful discrimination between participants with and without CVD.

3.6 Gender-Specific Models

Separate models for women and men showed that the overall structure of cardiovascular risk was similar across groups. Systolic blood pressure had nearly identical coefficients in women (0.05363) and men (0.05373), and high cholesterol remained important in both models. Age had a larger coefficient in women (0.05562) than in men (0.04423), whereas BMI had a larger coefficient in men (0.04132) than in women (0.02301). The protective association of physical activity was also stronger in men (-0.297) than in women (-0.217). Smoking showed a clearer adjusted association in the male model according to the source analysis.

These findings are more appropriately described as differences in magnitude than as entirely different risk structures. Both sexes shared the same principal clinical predictors, but some coefficients varied in strength.

Table 7: Comparison of selected gender-specific logistic coefficients

Predictor	Women	Men
Age	0.05562	0.04423
BMI	0.02301	0.04132
Systolic BP	0.05363	0.05373
Diastolic BP	0.01465	0.01813
High cholesterol	1.092	1.080
Physical activity	-0.217	-0.297

3.7 Model Performance

The female model achieved an AUC of 0.7936, the combined model 0.7910, and the male model 0.7871. The differences are small, so all three models can be characterized as having similar, acceptable-to-good discrimination. The results do not support a large performance advantage for one sex-specific model over the other.

logistic regression framework provides reliable predictive performance between gender-specific subsets, as well as in the combined population, making it a suitable tool for identifying people at elevated risk of cardiovascular disease.

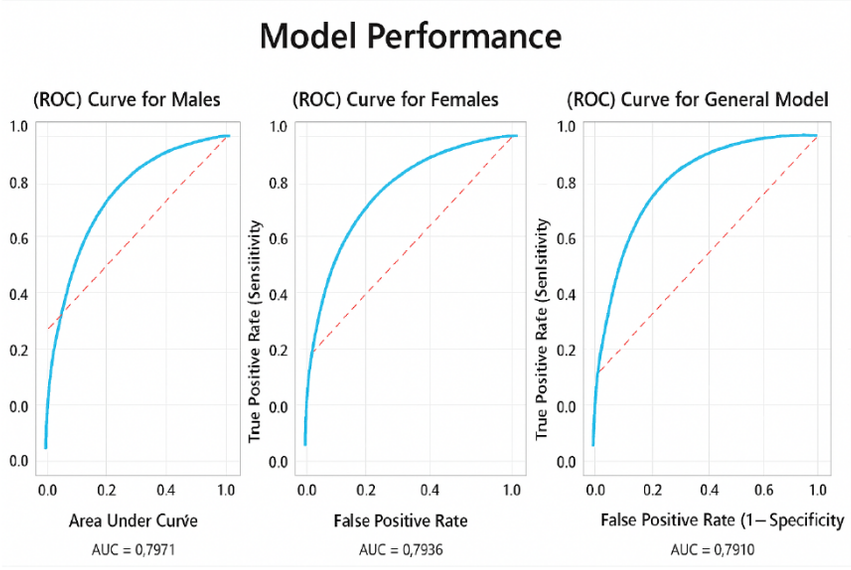


Figure 4.2: ROC curve for the logistic regression model (AUC reported by Minitab).

Figure 2: ROC curves for the male, female, and combined logistic regression models produced in Minitab.

4 Discussion

This project used several complementary statistical methods to examine cardiovascular risk in a large adult sample. The clearest and most consistent findings concerned age, blood pressure, BMI, and cholesterol. Individuals with CVD had higher mean BMI, metabolic categories were associated with meaningful differences in blood pressure and BMI, and high cholesterol was strongly overrepresented among participants with CVD. In the adjusted logistic model, age, BMI, systolic blood pressure, diastolic blood pressure, and cholesterol remained important predictors.

The gender-specific models add nuance rather than changing the overall conclusion. Systolic blood pressure and cholesterol were important in both groups. Age was more strongly associated with CVD among women in the fitted model, while BMI and the physical-activity coefficient were stronger among men. These results suggest that prevention can acknowledge sex-related differences in magnitude while still emphasizing the same core modifiable risk factors for both groups.

Several results require caution. In particular, inverse adjusted coefficients for smoking, alcohol use, and high glucose should not be interpreted as evidence that these exposures protect against CVD. The analysis is cross-sectional and relies on a public secondary dataset. Reverse causation, coding choices, clinical treatment after diagnosis, selection effects, or unmeasured confounding could produce such coefficients. This is an important reason to distinguish prediction from causal interpretation.

The project also demonstrates why very large samples require attention to effect size, not only p

values. With nearly 69,000 observations, even small differences can become highly statistically significant. The practical interpretation therefore depends on the size and direction of the estimated association, its clinical plausibility, and its stability in multivariable models.

4.1 Strengths and Limitations

A major strength is the large sample and the use of multiple statistical approaches, including nonparametric tests, multivariable regression, gender-stratified modeling, and ROC/AUC evaluation. The workflow is transparent and well suited to demonstrating applied analysis in Minitab.

The main limitations are the cross-sectional design, reliance on a public dataset with limited documentation, possible coding ambiguity in behavioral variables, lack of information on medication use and disease duration, and the absence of important socioeconomic and clinical covariates. Because the outcome represents existing disease rather than incident CVD, the logistic models should be interpreted as models of association and classification, not as prospective risk equations.

5 Conclusion

The analysis shows that cardiovascular disease status is strongly related to age, body mass index, blood pressure, and cholesterol in this sample. High cholesterol had one of the largest adjusted associations, and physical activity was associated with lower CVD odds. The combined logistic regression model achieved an AUC of 0.791, with very similar performance in the male and female models.

Gender-specific modeling showed differences in coefficient magnitude rather than fundamentally different cardiovascular risk profiles. Women had a stronger age coefficient, while men had stronger BMI and physical-activity coefficients; systolic blood pressure and cholesterol were consistently important in both groups. These findings support prevention strategies that emphasize weight management, blood-pressure control, lipid management, and physical activity while allowing for sex-specific tailoring where appropriate.

Future versions of this project would benefit from prospective data, clearer exposure coding, external validation, calibration analysis, and additional variables such as medication use, diabetes history, diet, socioeconomic status, and family history. Interaction terms could also be tested directly in a combined model rather than relying only on separate sex-stratified models.

Appendix: Model Equations

The fitted combined linear predictor from the source report was

$$\begin{aligned} Y' = & -11.491 + 0.05076(\text{age}) + 0.02787(\text{BMI}) \\ & + 0.053687(\text{SBP}) + 0.01593(\text{DBP}) \\ & + 0.3788(\text{chol}_1) + 1.0907(\text{chol}_2) \\ & + 0.0185(\text{gluc}_1) - 0.3355(\text{gluc}_2) \\ & - 0.1383(\text{smoke}) - 0.2077(\text{alcohol}) \\ & - 0.2304(\text{active}) + 0.0315(\text{male}), \end{aligned}$$

with $P(\text{CVD} = 1) = \exp(Y')/[1 + \exp(Y')]$.

References

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